Activity - 1

1) Medical Image Enhancement :

i) I recommend the Power-Law (Gamma) transformation with $\gamma < 1$ to improve the visibilty of dark tissues in the X-ray image.

ii) This transformation increases the brightness of darker pixels and improve the visibility.

iii) The dark issues become brighter and more visible, while the bone structures remain visible. It improves the overall appearance of the X-ray image.

2) Night - Time Surveillance :

i) I recommend the logarithmic transformation because most pixel value are concentrated between 0 & 40, making the image very dark.

ii) It expands low-intensity pixel values and reveal details in dark regions. Negative transformation reverse the brightness values but does not specifically enhance dark details in the same way.

iii) The parking lot appears brighter and object such as cars, people and shadow become easier to identify.

3) Fingerprint Recognition:

i) I recommend Contrast Stretching transformation because the fingerprint image has low contrast between ridges and valleys.

ii) Contrast Stretching - Increase the difference between dark and light pixel values. It make fingerprint ridges and valleys clear, helping the fingerprint recognition system.

iii) Log transformation brightens darker regions and compresses higher intensity values. However, it may not improve the contrast between ridges and valleys as effectively as contrast stretching.